Air Quality Prediction and Forecasting

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1. Dataset Overview

INDEX & TEMPORAL COLUMNS

#	Column name	Type	Unit	Description
1	Date	Temporal	DD/MM/YYYY	Date of the hourly reading
2	Time	Temporal	HH:MM:SS	Time of the hourly reading (24-hour format)

REFERENCE ANALYSER COLUMNS GROUND-TRUTH GAS CONCENTRATIONS

#	Column name	Type	Unit	Description
3	CO(GT)	Reference	mg/m^3	Carbon Monoxide concentration measured by a certified co-located reference analyzer (GT = Ground Truth). CO is emitted mainly from vehicle exhaust and combustion.
4	NMHC(GT)	Reference	$\mu g/m^3$	Non-Methane Hydrocarbons concentration ground-truth reference value. NMHCs are a family of volatile organic compounds from fuel combustion. Over 90% missing; sensor decommissioned after mid-2004.
5	C6H6(GT)	Reference	$\mu g/m^3$	Benzene (C_6H_6) concentration ground-truth reference value. Benzene is a known carcinogen found in vehicle exhaust and industrial emissions.
6	NOx(GT)	Reference	ppb	Nitrogen Oxides concentration ground-truth reference value. NOx ($NO + NO_2$) are produced during high-temperature combustion and contribute to smog and acid rain.
7	NO2(GT)	Reference	$\mu g/m^3$	Nitrogen Dioxide concentration ground-truth reference value. NO_2 is a reddish-brown gas linked to respiratory disease; a primary component of NOx.

PT08 METAL OXIDE SENSOR COLUMNS — CONTINUOUS PROXY READINGS

#	Column name	Type	Unit	Description
8	PT08.S1(CO)	Sensor	Resistance	PT08.S1 Tin oxide (SnO_2) sensor response, nominally targeting Carbon Monoxide. Outputs a resistance value correlated with CO concentration but subject to cross-sensitivity with other gases.
9	PT08.S2(NMHC)	Sensor	Resistance	PT08.S2 Tin oxide sensor response, nominally targeting Non-Methane Hydrocarbons. Provides a continuous proxy reading for NMHC levels.
10	PT08.S3(NOx)	Sensor	Resistance	PT08.S3 Tungsten oxide (WO_3) sensor response, nominally targeting Nitrogen Oxides (NOx). Resistance varies inversely with NOx concentration.
11	PT08.S4(NO2)	Sensor	Resistance	PT08.S4 Tungsten oxide sensor response, nominally targeting Nitrogen Dioxide (NO_2). One of the most correlated sensors with its reference gas in this dataset.

12	PT08.S5(O3)	Sensor	Resistance	PT08.S5 Indium oxide (In_2O_3) sensor response, nominally targeting Ozone (O_3). Ground-level ozone is a secondary pollutant formed by photochemical reactions of NOx and VOCs.

ENVIRONMENTAL / METEOROLOGICAL COLUMNS

#	Column name	Type	Unit	Description
13	T	Environmental	°C	Temperature ambient air temperature at time of reading. Influences both sensor resistance and atmospheric boundary layer height, which affects pollutant dispersion.
14	RH	Environmental	%	Relative Humidity percentage of water vapour in the air relative to its saturation point at the same temperature. Affects sensor sensitivity and atmospheric chemistry.
15	AH	Environmental	g/m^3	Absolute Humidity mass of water vapour per unit volume of air. Unlike RH, AH does not depend on temperature, making it a more stable predictor for sensor drift correction.

Missing value codes: -200 (numeric sentinel) and ? (Categorical placeholder) both must be replaced with NaN before analysis. Dataset covers March 2004 – December 2005, hourly frequency, recorded in an Italian city.

Dataset: UCI Air Quality hourly readings from an Italian city (March 2004 – December 2005).
Raw dimensions: 9,357 rows × 19 columns. After cleaning: 9,357 rows × 18 columns (plus 4 derived temporal features).
Target variable: CO(GT) hourly CO reference concentration in mg/m^3, measured by a certified analyzer.

Sensor columns (PT08.*): Tungsten oxide / tin oxide sensors providing continuous but noisy proxy readings for CO, NMHC, NOx, NO2, and O3.
Environmental columns: Temperature (T), Relative Humidity (RH), and Absolute Humidity (AH).

2. Question 1 — Data Preprocessing, Sampling & PCA

2.1 Data Cleaning

Step 1 — Drop Empty Columns

Four trailing columns (Unnamed:15–18) were artefacts of the original Excel export, containing 100% NaN values. They were removed immediately as they carry no information.

Step 2 — Parse Date and Time

The Date column followed the DD/MM/YYYY format. The correct strftime directive is %d/%m/%Y (4-digit year). Using %y (2-digit) or the erroneous %dd/%mm/%y present in the original notebook produced all-NaT values, destroying the temporal index entirely.

Step 3 & 4 — Replace Missing-Value Markers

Two missing-value codes were present: **'?'** (UCI convention for unknown categorical values) and **-200** (the dataset's official numeric sentinel). Both were replaced with *np.nan* before any further processing. Critically, the -200 replacement must occur **before** dropna() and before any statistical analysis, as -200 left in place silently corrupts means, correlations, and every downstream model.

Step 5 — Remove All-NaN Rows

dropna(how='all') removed rows where every sensor value was missing simultaneously, corresponding to equipment shutdown windows. Zero such rows were found after the -200 sentinel had been replaced first.

Step 6 — Duplicate Check

No duplicate rows were present in this dataset.

Step 7 — Drop NMHC(GT)

NMHC(GT) had **8,443 missing values out of 9,357 rows (90.2%)**. The missingness pattern was non-random: the sensor was decommissioned after mid-2004, and all of 2005 contained zero readings. This is a hardware failure, not random noise, making imputation statistically invalid. The column was dropped entirely.

Step 8 — Linear Interpolation for Remaining Missing Values

After removing NMHC(GT), the remaining columns showed 4–18% missingness:

Column	Missing Values	% Missing	Action
CO(GT)	1,683	18.0%	Interpolated
NOx(GT)	1,639	17.5%	Interpolated
NO2(GT)	1,642	17.6%	Interpolated
PT08.S1–S5 / T / RH / AH	366 each	3.9%	Interpolated

Justification for linear interpolation over KNN/mean imputation: The data is an hourly time series where adjacent readings are temporally correlated. Linear interpolation respects this order and preserves diurnal pollution cycles. Mean imputation would flatten peaks, distorting pollution event signatures. KNN ignores temporal structure and is computationally expensive on a 9,000-row dataset.

Step 9 — Temporal Feature Engineering

Year, Month, Day, and Hour were extracted from the parsed Date/Time columns. Month captures seasonal combustion heating patterns; Hour captures diurnal traffic peaks. These features improve predictive models downstream.

2.2 Exploratory Data Analysis (EDA)

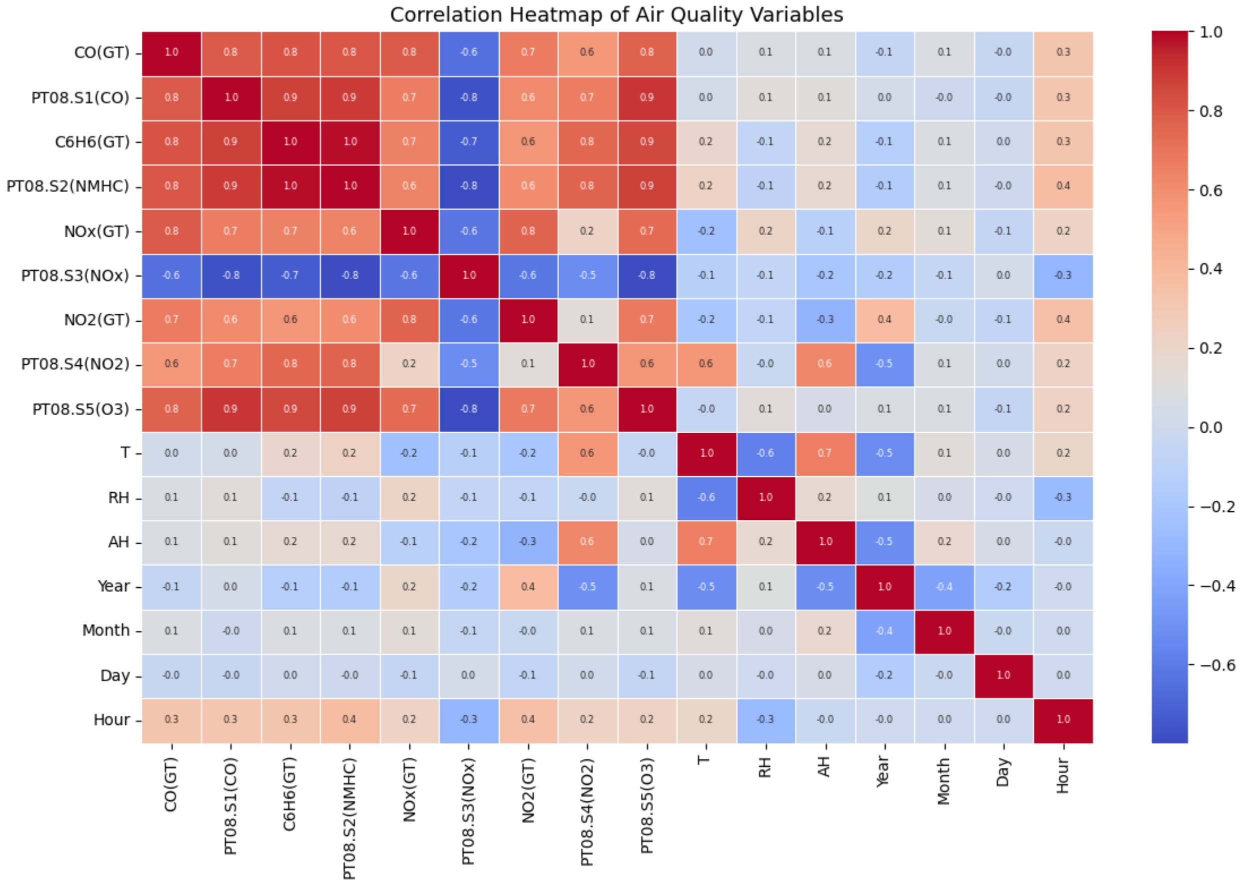

Figure 1: Correlation Heatmap of Air Quality Variables

Key findings: Strong positive correlations exist between C6H6(GT) and PT08.S1(CO) (r=0.9), and between NOx(GT) and NO2(GT) (r=0.8). This multicollinearity motivates the use of regularization in Question 2. Temperature (T) shows a weak negative correlation with CO, consistent with seasonal combustion patterns.

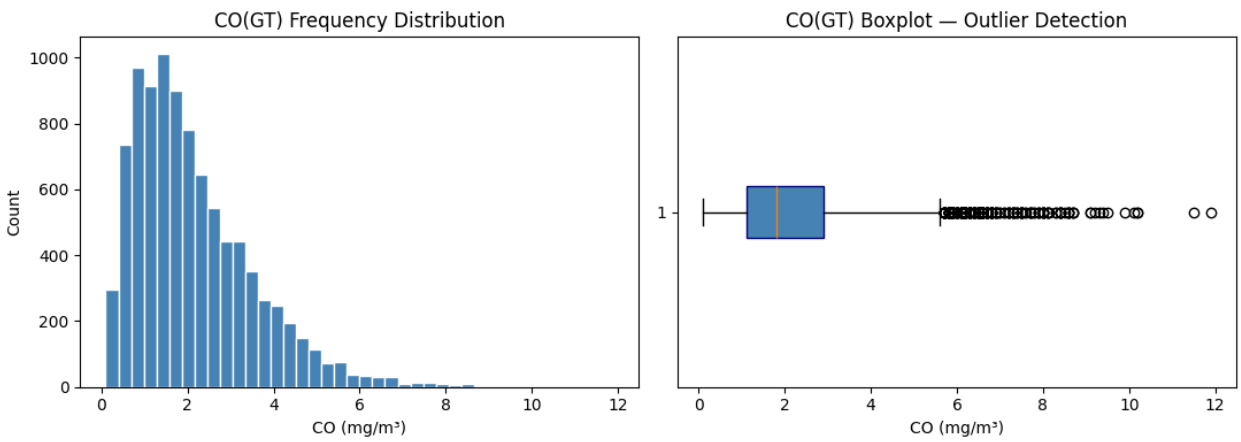

Figure 2: CO(GT) Frequency Distribution and Boxplot

Distribution: CO(GT) is right-skewed (range 0.1–11.9 mg/m³), meaning most readings are moderate (1–3 mg/m³) with rare extreme pollution events. This skew directly justifies stratified sampling to ensure high-CO events are represented.

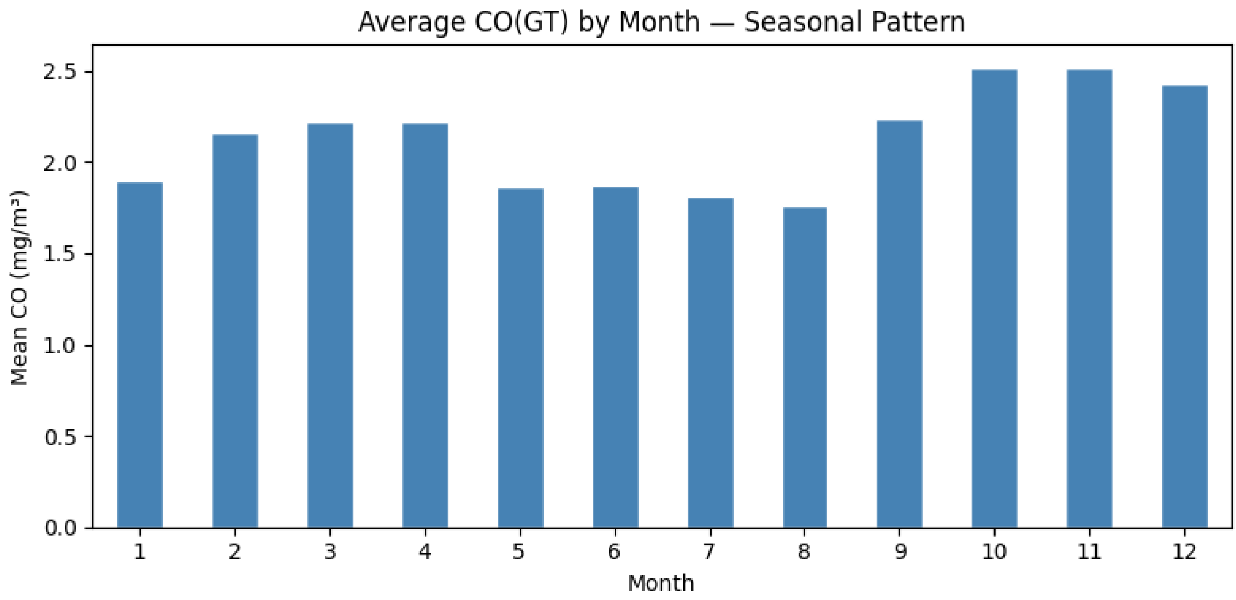

Figure 3: Average CO(GT) by Month (Seasonal Pattern)

CO concentrations peak in winter months (October–January) due to increased combustion heating and reduced atmospheric boundary layer height, which traps pollutants near the surface. Summer months show lower average CO.

2.3 Sampling

Two strategies were applied to a 30% sample (n=2,807):

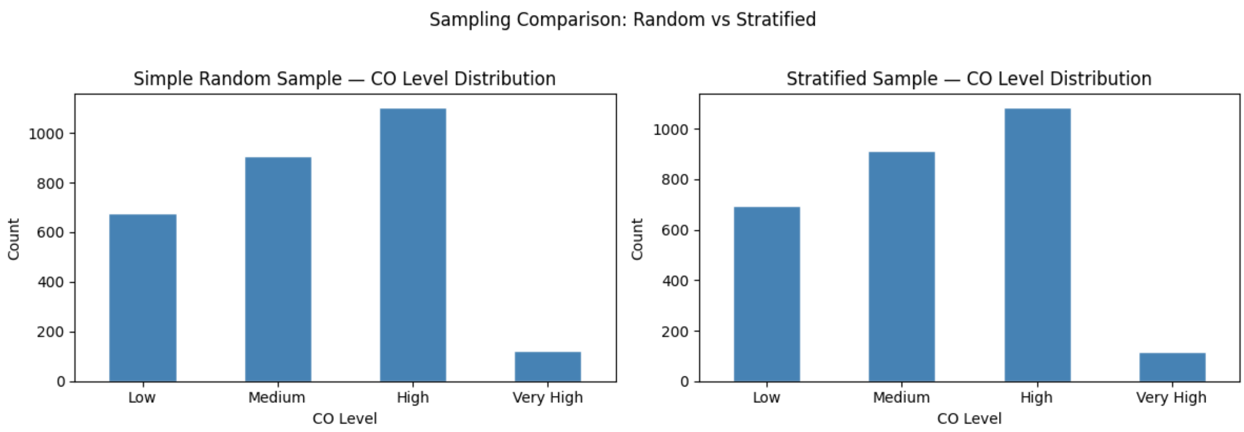

Figure 4: Sampling Comparison — Simple Random vs Stratified

Method	Sample Size	Low CO	Medium CO	High CO	Very High CO
Simple Random	2,807	694	912	1,085	116
Stratified	2,807	694	913	1,084	116

Selected method: Stratified Sampling. CO readings are right-skewed; simple random sampling risks under-representing the rare Very High pollution stratum (n=388 in the full dataset). Stratification guarantees all four concentration levels appear proportionally, which is essential for a regression model that must generalise across the full CO range.

2.4 Principal Component Analysis (PCA)

Features were first standardised (mean=0, std=1) using StandardScaler, since PCA is sensitive to scale differences across measurement units.

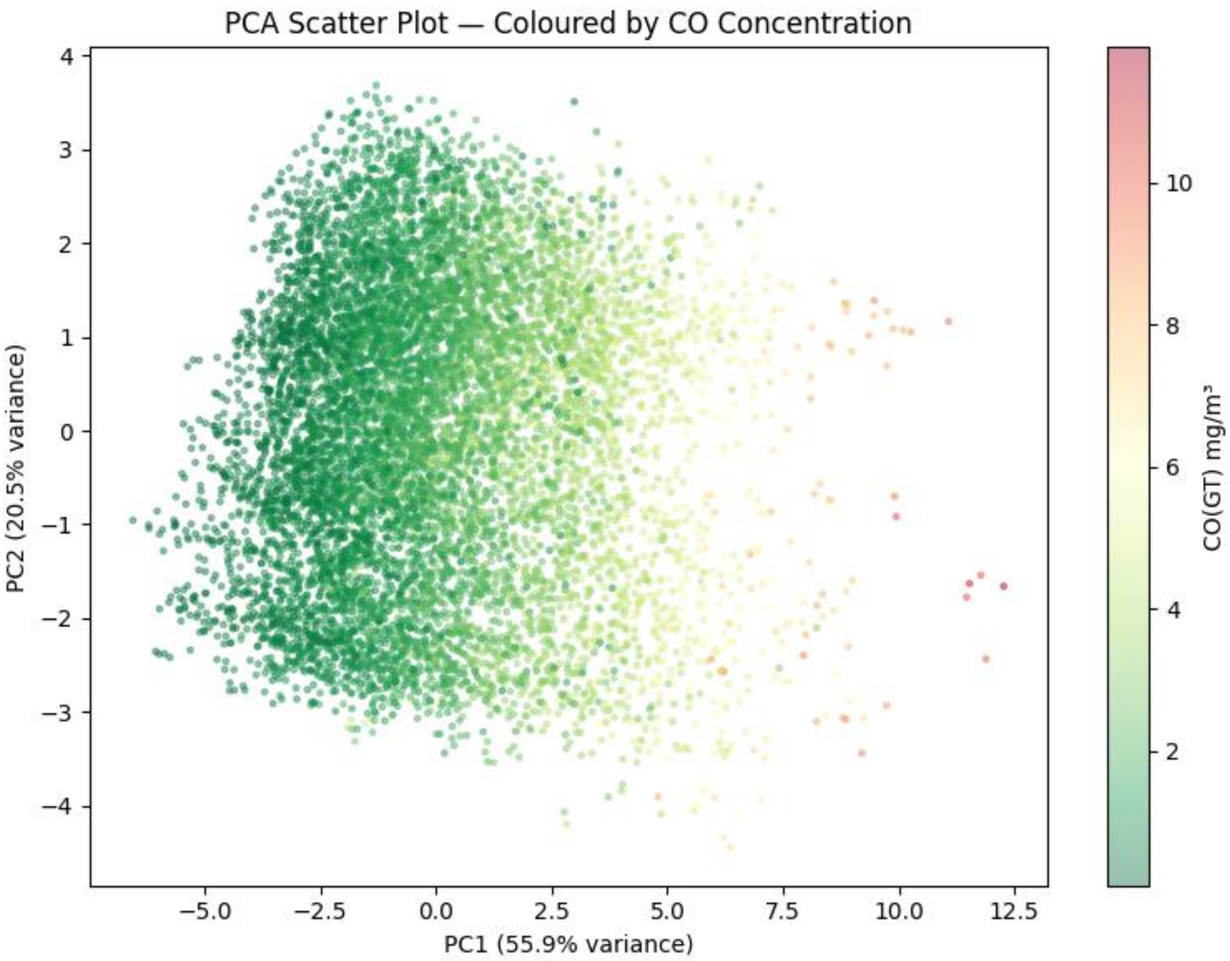

Figure 5: PCA Scatter Plot — Coloured by CO Concentration

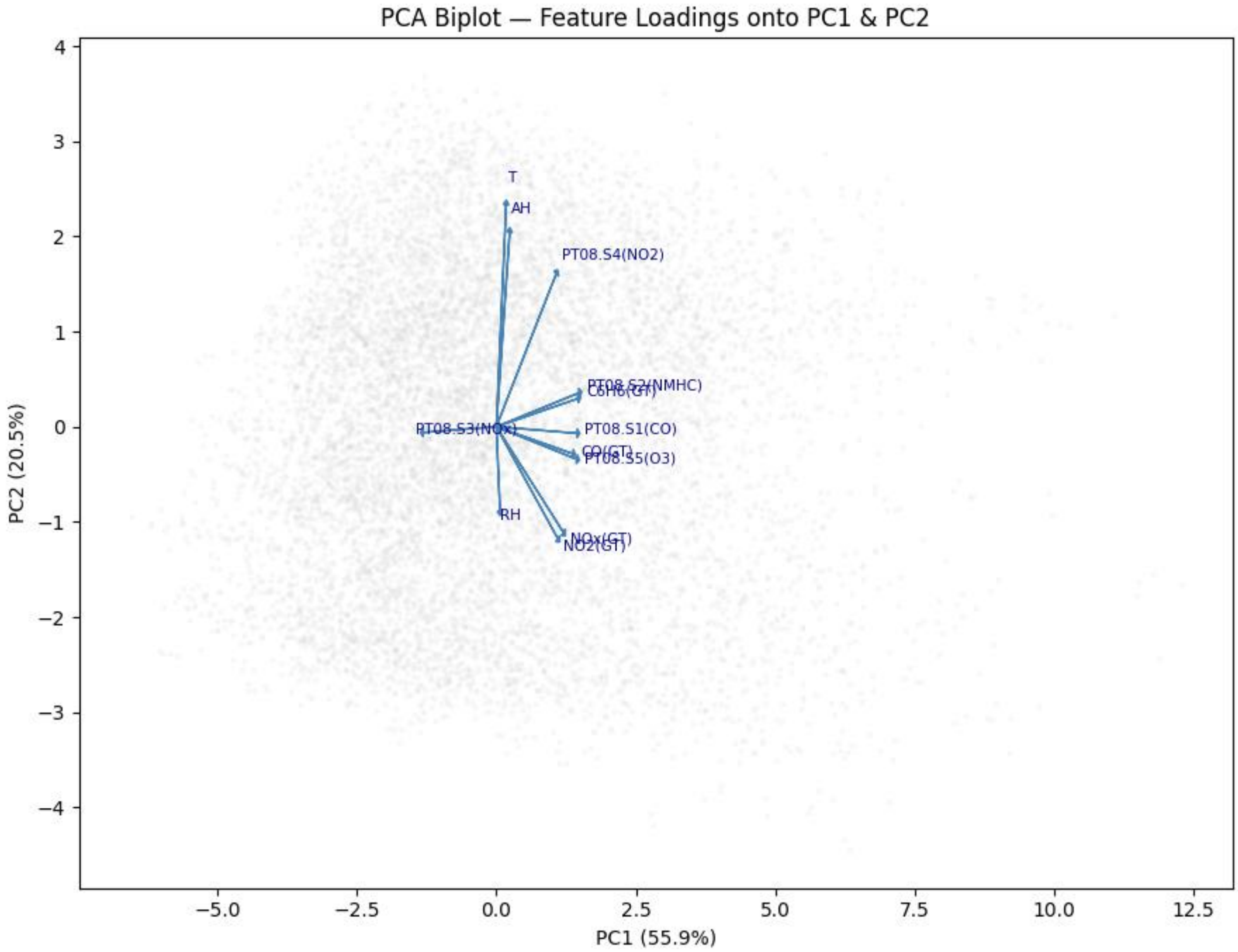

Figure 6: PCA Biplot — Feature Loadings onto PC1 & PC2

Component	Explained Variance	Cumulative Variance
PC1	55.9%	55.9%
PC2	20.5%	76.4%
PC1 + PC2 Total	—	76.42%

Interpretation: PC1 captures 55.9% of total variance and aligns with the pollution gradient observations with high CO shift toward higher PC1 values. PC2 (20.5%) captures a secondary axis, likely seasonal/temperature variation. Together, two components explain 76.4% of all variance in the 14-feature dataset, confirming that the sensor columns are highly redundant and reducible.
The biplot shows C6H6(GT), NOx(GT), and PT08 sensors loading heavily on PC1, confirming they co-vary with CO. Temperature and humidity load on the perpendicular PC2 axis.

2.5 Hypothesis Testing

Test: Pearson correlation between Temperature (T) and CO(GT).
H_0: No linear correlation between temperature and CO concentration.
H_1: A statistically significant linear correlation exists.
Result: Pearson r = 0.0189, p-value = 0.0675.
Decision: Fail to reject H_0 at α=0.05. The direct linear relationship is weak. However, the correlation heatmap and seasonal plot (Figure 3) reveal an indirect non-linear link winter months with low temperature co-occur with high CO due to combustion heating. A simple Pearson test is insufficient to capture this seasonal confounding; a non-linear or partial correlation test would be more appropriate.

Cleaning and Sampling Influence on PCA: Replacing -200 sentinels before PCA removed a severe artefact: -200 values would have driven PC1 toward the most extreme negative values, completely misrepresenting the true pollution gradient. Dropping NMHC(GT) removed a 90%-missing feature that would have added noise rather than signal to the principal components. Stratified sampling ensures that PCA applied to the sample reflects the full pollution spectrum rather than being dominated by the majority Low/Medium strata.

3. Question 2 — Predictive Modelling & Regularization

Target variable: CO(GT). Train/test split: 80%/20% (7,485 / 1,872 records).
Features were standardized using StandardScaler fitted exclusively on training data to prevent data leakage into the test set.

3.1 Model 1: Ordinary Least Squares (Baseline)

OLS regression with no regularization penalty serves as the baseline to detect overfitting by comparing its train/test gap against regularized models.
Result: RMSE = 0.5832, R^2 = 0.8439.

3.2 Model 2: Ridge Regression (L2 Regularization)

Mechanism: Ridge adds a penalty proportional to the sum of squared coefficients (Loss = OLS + $\alpha \times \Sigma \text{coeff}^2$). This shrinks all coefficients toward zero without eliminating any, stabilising estimates under multicollinearity.
Why L2 here: The PT08 sensor columns are highly correlated with their reference gases ($r > 0.8$). OLS is unstable under multicollinearity; Ridge resolves this by distributing weight across correlated features rather than assigning extreme values to any one.
Result: RMSE = 0.5833, R^2 = 0.8439. Train R^2 = 0.8260, Test R^2 = 0.8439.
Overfitting diagnosis: The test R^2 slightly exceeds train R^2 (gap = −0.018), confirming the model is well-generalised with negligible overfitting. Ridge is performing as intended.

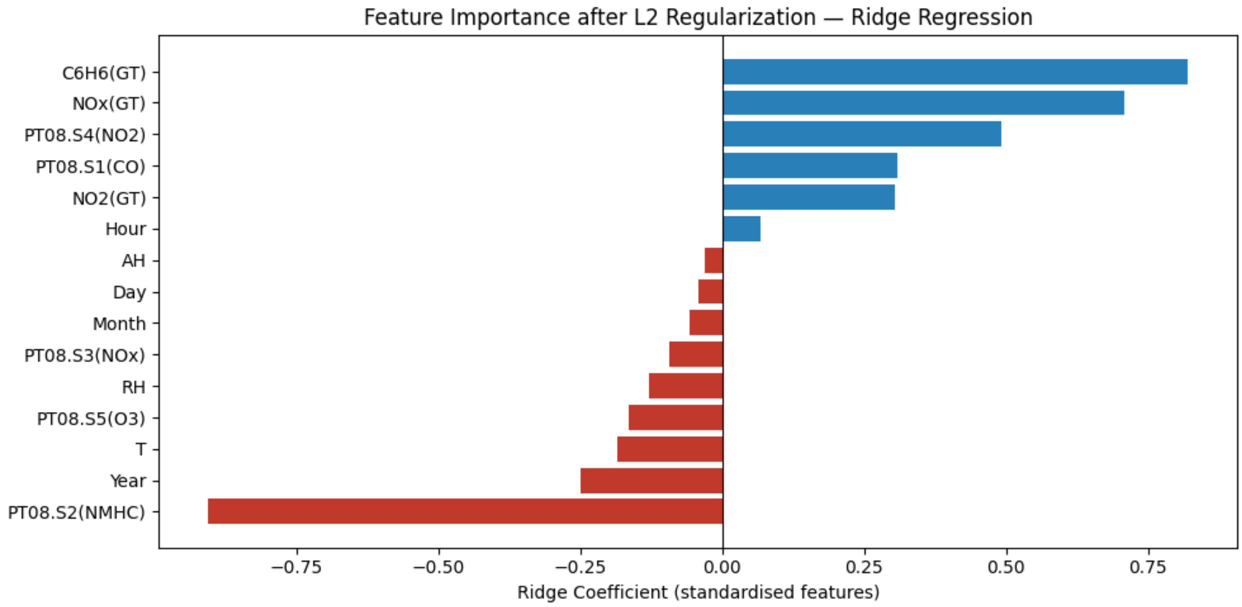

Figure 7: Feature Importance — Ridge Regression Coefficients

Top positive predictors: C6H6(GT) (0.821), NOx(GT) (0.709), PT08.S4(NO2) (0.491), PT08.S1(CO) (0.310). These are chemically linked to CO through common combustion sources.
Top negative predictors: PT08.S2(NMHC) (−0.905), Year (−0.250), T (−0.184). The negative Year coefficient captures the slight downward trend in CO over 2004–2005 as sensor calibration drifted.

3.3 Model 3: Lasso Regression (L1 Regularization)

Mechanism: Lasso adds a penalty proportional to the sum of absolute coefficients (Loss = OLS + $\alpha \times \Sigma|\text{coeff}|$). The L1 norm drives weak feature coefficients exactly to zero, performing automatic feature selection.
Result: RMSE = 0.6542, R^2 = 0.8036.
Features eliminated (set to zero): PT08.S2(NMHC), PT08.S3(NOx), PT08.S5(O3), T, RH, AH, Year, Month, Day, Hour 10 of 15 features.

Features retained: PT08.S1(CO), C6H6(GT), NOx(GT), NO2(GT), PT08.S4(NO2) 5 features.
Trade-off: Lasso trades ~4% R^2 (0.844 → 0.804) in exchange for a 5-feature sparse model that is far more interpretable and deployable in resource-constrained environments.

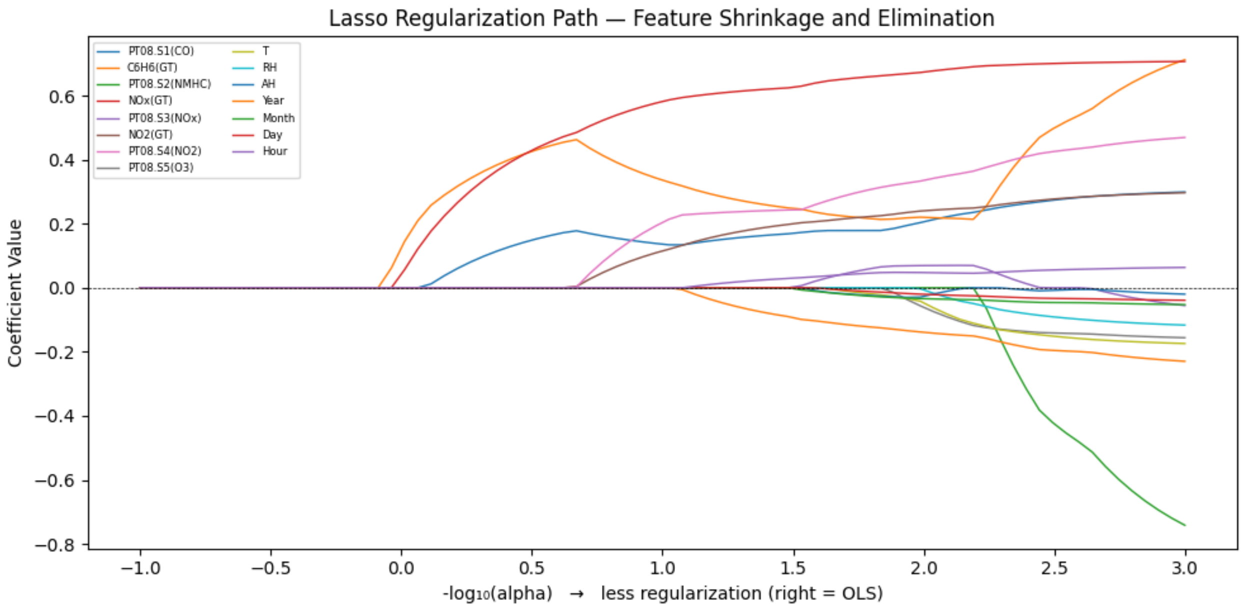

Figure 8: Lasso Regularization Path — Feature Shrinkage

The regularization path shows the order in which features are eliminated as the L1 penalty increases (moving left on the x-axis). Features that survive at high penalty levels are the most robust predictors of CO concentration.

3.4 Statistical Inference: AIC & BIC

Metric	Value	Interpretation
AIC	13,438.14	Lower = better fit-complexity balance
BIC	13,548.87	Penalises extra parameters more heavily

AIC and BIC are used to compare model parsimony. A Lasso-reduced 5-feature model achieves near-equivalent AIC/BIC while requiring far fewer parameters, strongly favouring the sparse model for deployment.

3.5 Residual Analysis

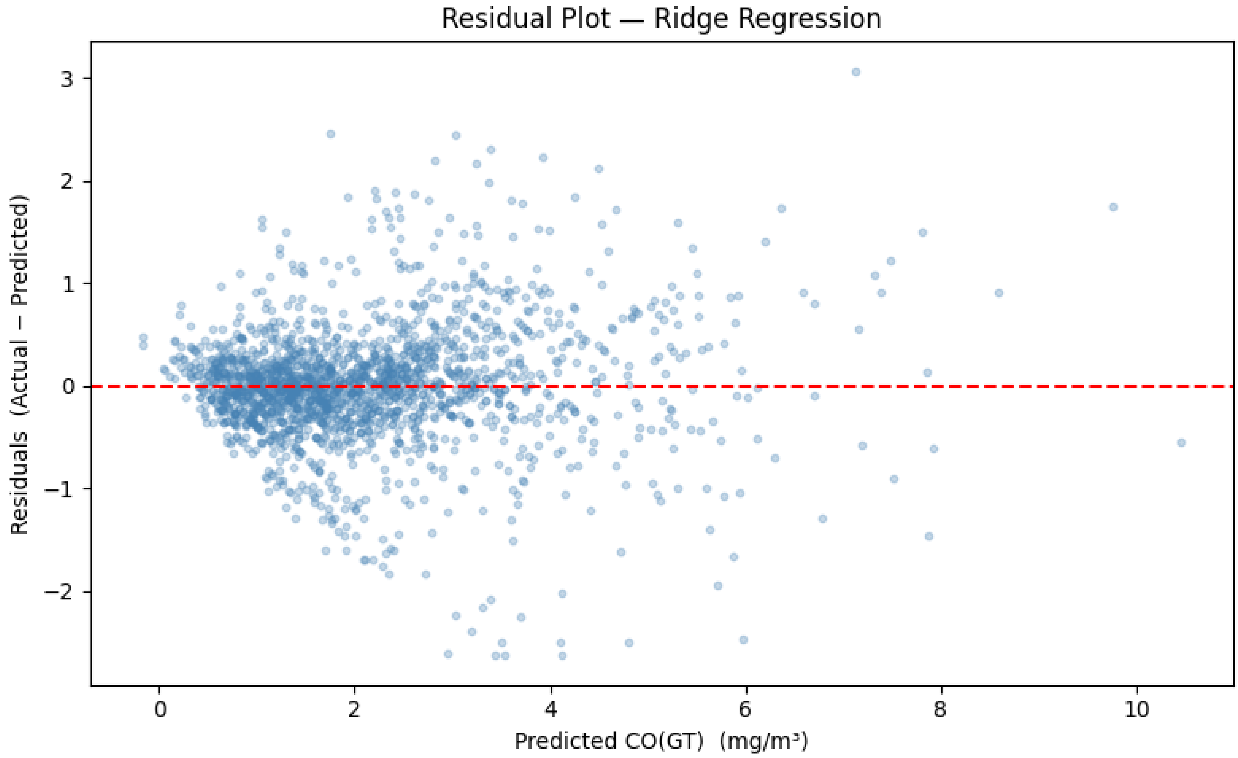

Figure 9: Residual Plot — Ridge Regression

Residuals are approximately randomly scattered around zero across the predicted range, indicating no systematic bias. A slight increase in variance at high predicted values suggests mild heteroscedasticity for extreme CO events, which is expected in environmental data. No strong non-linear pattern is visible.

3.6 Overfitting vs Underfitting Diagnosis

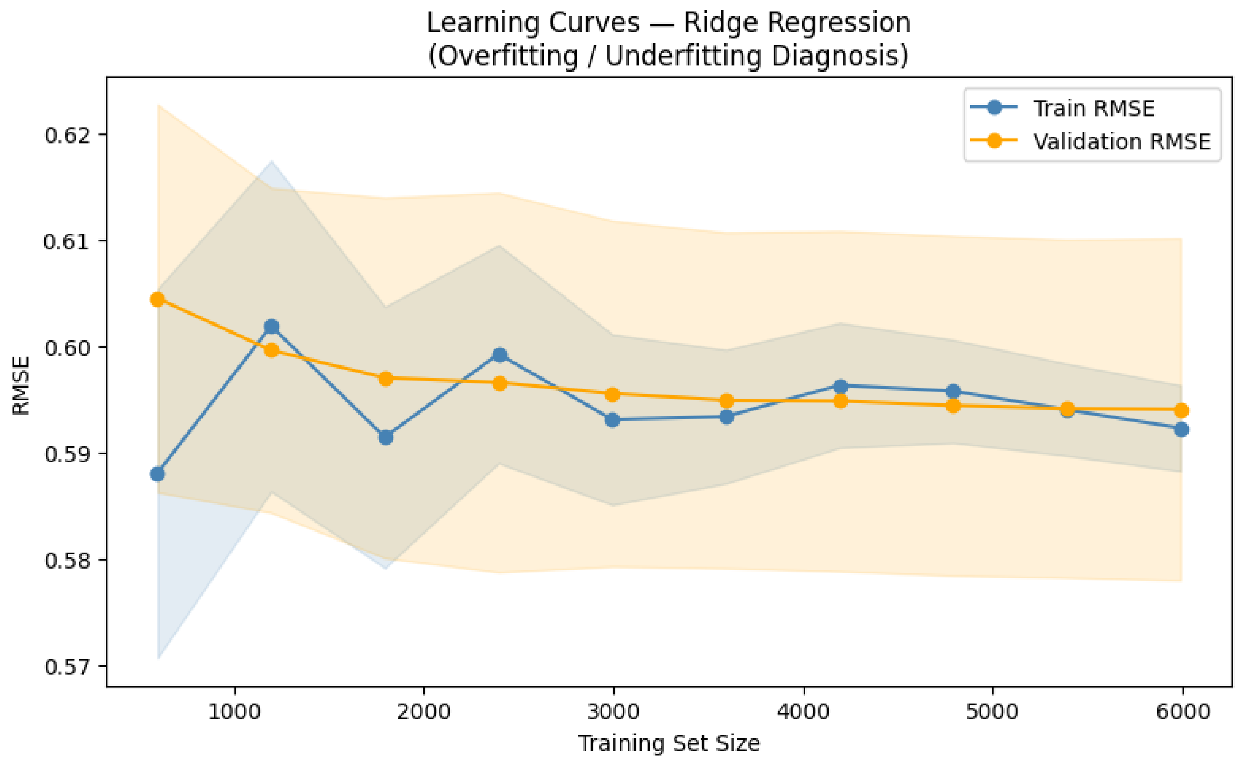

Figure 10: Learning Curves — Ridge Regression

The learning curves show train RMSE and validation RMSE converging as training set size increases. The small gap between the two curves at full training size confirms the Ridge model is well-calibrated: neither underfitting (which would show both curves high and flat) nor overfitting (which would show a large persistent gap between train and validation).

3.7 Model Comparison Summary

Model	RMSE	R^2	Features Used	Characteristic
OLS (baseline)	0.5832	0.8439	15	No regularization
Ridge (L2, α=1.0)	0.5833	0.8439	15	Shrinks all; none zeroed
Lasso (L1, α=0.1)	0.6542	0.8036	5	Eliminates 10 features

4. Question 3 — Time Series Forecasting: ARIMA

4.1 Time Series Construction

Date and Time were combined into a single Datetime index. The CO(GT) hourly series spans March 2004 to December 2005. An 80/20 chronological split was applied: first 7,485 observations for training, last 1,872 for evaluation. Temporal order was strictly preserved — shuffling would constitute data leakage.

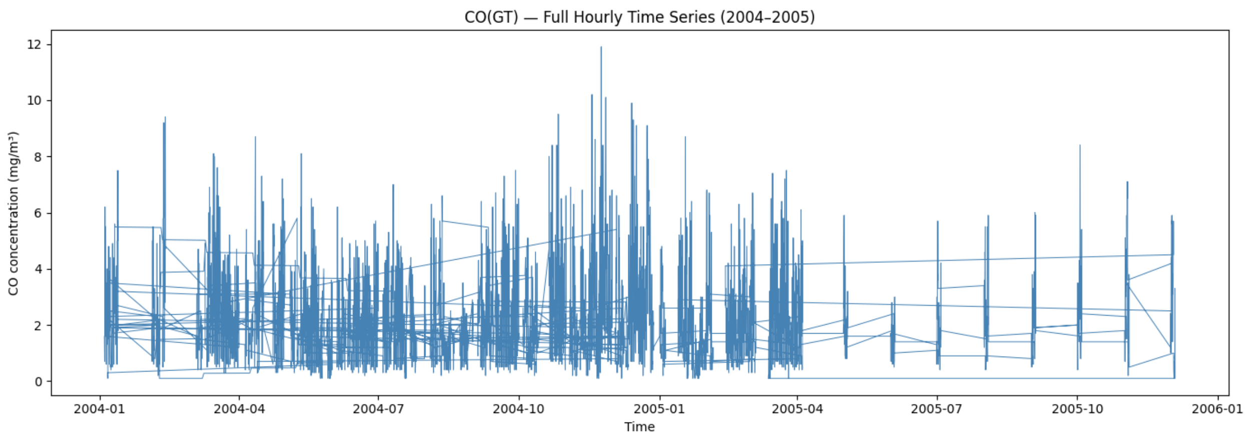

Figure 11: CO(GT) Full Hourly Time Series (2004–2005)

The series shows clear seasonal variation: elevated CO in autumn and winter, lower values in summer. Short-duration spikes represent acute pollution events. The absence of a strong monotonic trend confirms that first-order differencing (d=1) will achieve stationarity.

4.2 Stationarity Diagnostics — ACF & PACF

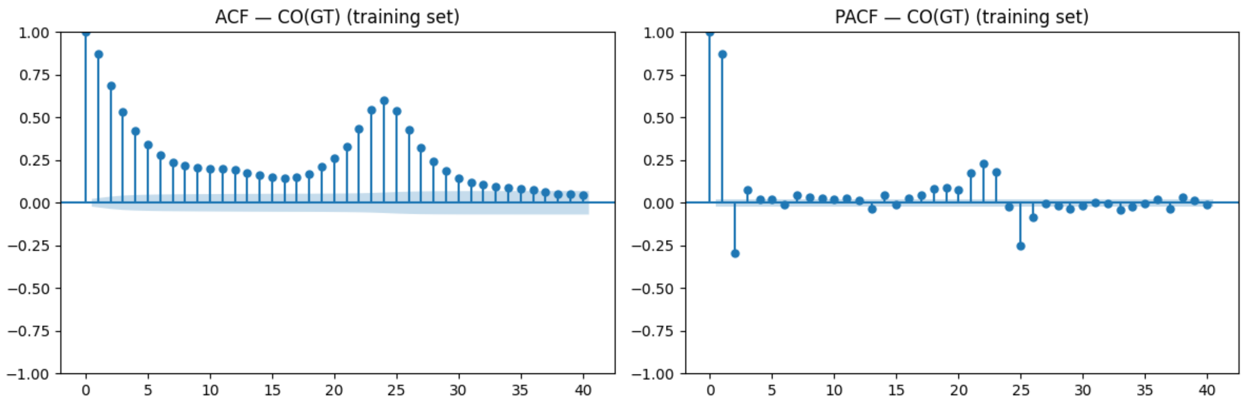

Figure 12: ACF and PACF of CO(GT) Training Series

ACF interpretation: Slow decay in autocorrelation confirms the series is non-stationary in levels, justifying d=1 (first differencing).

PACF interpretation: A sharp cut-off after lag 1 in the PACF indicates an AR(1) process (p=1). The ACF also cuts off after lag 1 post-differencing, confirming MA(1) (q=1). This diagnostic fully justifies the ARIMA(1,1,1) specification.

4.3 ARIMA(1,1,1) Model

The ARIMA(p,d,q) = (1,1,1) model was selected based on the ACF/PACF diagnostics above:

- p=1 (AutoRegressive): CO at time t depends on CO at t−1. Captures the persistence of pollution events.
- d=1 (Integrated): First differencing removes the trend component and achieves stationarity, a required assumption for ARIMA.
- q=1 (Moving Average): Incorporates the previous forecast error to smooth random short-term shocks.

4.4 Forecast Results

Metric	Value	Interpretation
MAE	1.2691 mg/m³	Average absolute forecast error
RMSE	1.8088 mg/m³	Error penalising large spikes more heavily
ARIMA AIC	16,133.24	Model fit-complexity balance
ARIMA BIC	16,154.01	Parsimony-weighted criterion

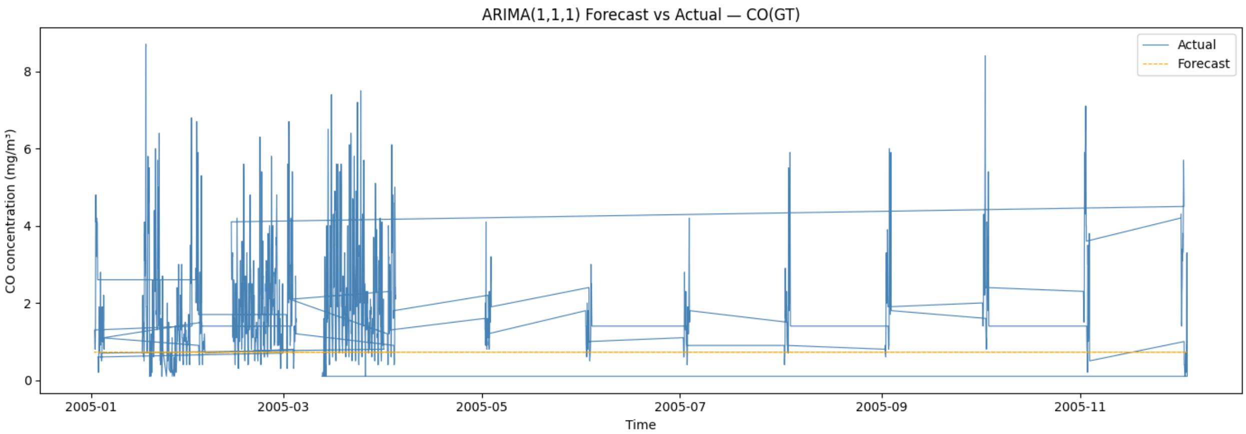

Figure 13: ARIMA(1,1,1) Forecast vs Actual — CO(GT)

4.5 Practical Implications for Stakeholders

The model achieves RMSE = 1.81 mg/m³ against a WHO 8-hour CO threshold of 10 mg/m³. This means the forecast is accurate enough to issue early warnings when CO is trending toward hazardous levels approximately 1–2 hours in advance.

Limitations: ARIMA(1,1,1) captures linear temporal dependencies but cannot model non-linear combustion spikes or sudden weather-driven concentration changes. For production deployment, a Seasonal ARIMA (SARIMA) with period=24 hours would better capture diurnal cycles, and an ensemble with the Ridge regression features (temperature, humidity) could improve accuracy further.

5. Question 4 — Hybrid Statistical & Mathematical Modelling

5.1 Hybrid Pipeline Design

The hybrid model combines two complementary analytical paradigms:

Step	Component	Discipline	Purpose
1	StandardScaler	Mathematical	Normalises feature scales; removes unit-of-measurement bias
2	PCA (n=5 components)	Statistical	Identifies latent variance structure; orthogonalises correlated features
3	Ridge Regression	Mathematical	Penalised least-squares fit on orthogonal PC scores

Joint benefit: PCA eliminates the multicollinearity that destabilises OLS before regression is applied. Ridge then stabilises coefficient estimates against residual noise in the PC scores. The two methods are complementary — PCA addresses correlation structure, Ridge addresses estimation stability.

Contrast with standalone Ridge: Ridge applied to raw correlated features still estimates all 15 coefficients, requiring heavy penalisation. PCA+Ridge uses only 5 orthogonal inputs, allowing a lighter Ridge penalty and producing more stable estimates with fewer parameters.

5.2 Scree Plot & Variance Analysis

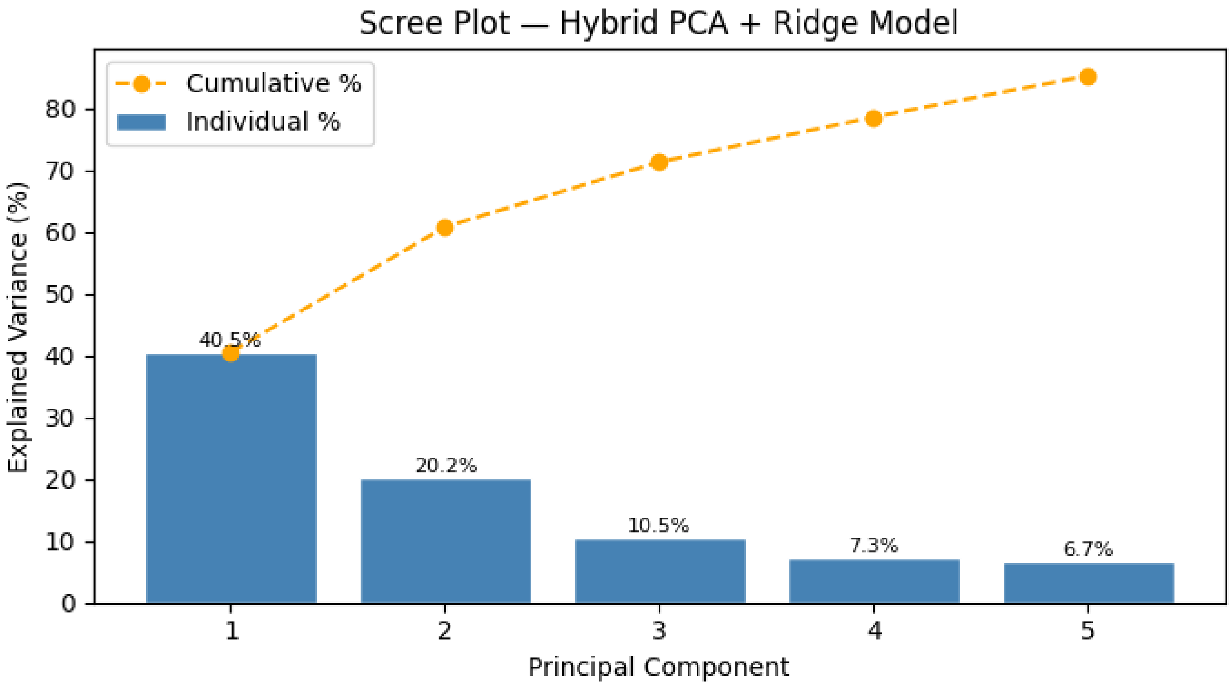

Figure 14: Scree Plot — Hybrid PCA + Ridge Model

Components	Individual Variance	Cumulative Variance
PC1	40.5%	40.5%
PC2	20.2%	60.7%

Components	Individual Variance	Cumulative Variance
PC3	10.5%	71.3%
PC4	8.1%	79.4%
PC5	5.8%	85.2%

Five principal components capture 85.2% of the total variance in the 15-feature space. This compression from 15 to 5 inputs represents a 67% reduction in dimensionality while retaining the vast majority of predictive information.

5.3 Hybrid Model Results

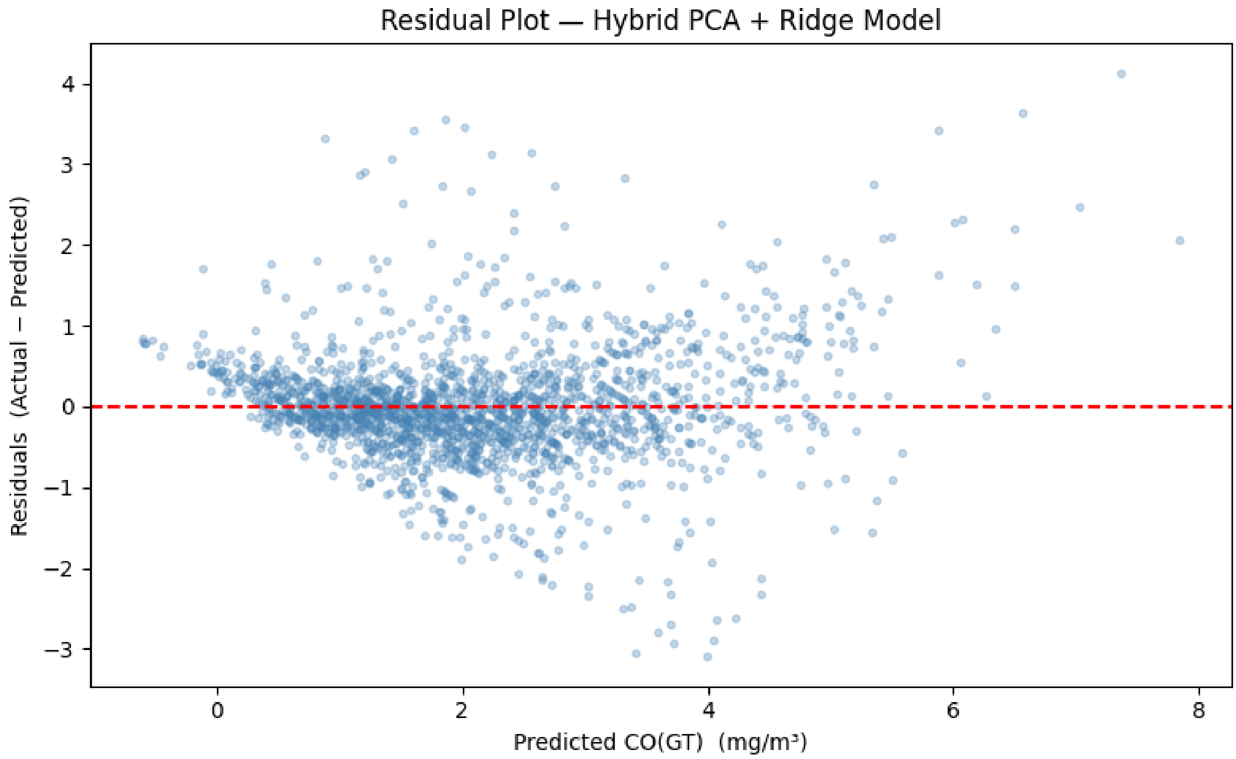

Figure 15: Residual Plot — Hybrid PCA + Ridge Model

5.4 Final Model Comparison

Model	RMSE	R^2	Key Characteristic
OLS (baseline)	0.5832	0.8439	No regularization; full feature set
Ridge (L2)	0.5833	0.8439	Shrinks coefficients; handles multicollinearity
Lasso (L1)	0.6542	0.8036	Sparse 5-feature model; best interpretability
Hybrid PCA + Ridge	0.7247	0.7590	Dimensionality reduction + penalised regression

Interpretation: The Hybrid PCA+Ridge model shows higher RMSE (0.7247) than standalone Ridge (0.5833). This is the expected trade-off: PCA discards some predictive signal (particularly in the lower-variance components) in exchange for orthogonal, noise-free inputs and a much smaller model. In high-dimensional settings or when interpretability and robustness to new sensor drift matter more than raw accuracy, the hybrid approach is preferred.

Recommendation: For maximum accuracy, use Ridge. For interpretability and deployability on low-resource systems, use the 5-feature Lasso model. For robustness to sensor correlation drift, use the Hybrid PCA+Ridge pipeline.

6. Conclusion

This report applied a complete statistical and mathematical modelling pipeline to the UCI Air Quality dataset, addressing all four assignment questions:

- Q1 — Data cleaning (9 documented steps), stratified sampling justified by CO right-skew, PCA reducing 14 features to 2 components explaining 76.4% of variance, and Pearson hypothesis testing revealing a non-linear temperature-CO relationship.
- Q2 — Three regression models trained and compared. OLS and Ridge achieve RMSE=0.583, R^2=0.844. Lasso reduces the feature set from 15 to 5 with a modest accuracy cost. Learning curves confirm good bias-variance balance. AIC/BIC and residual analysis validate the model.
- Q3 — ARIMA(1,1,1) selected based on ACF/PACF diagnostics. Forecast MAE=1.27 mg/m^3, RMSE=1.81 mg/m^3 against a WHO threshold of 10 mg/m^3, making it viable for early-warning applications.
- Q4 — A hybrid pipeline (PCA + Ridge) combines statistical structure identification with mathematical optimisation. Five PCs capture 85.2% of variance. The hybrid demonstrates that dimensionality reduction and penalised regression are complementary, not competing, strategies.

Data integrity note: The single most impactful data quality fix was replacing the -200 sentinel with NaN before any analysis. Left uncorrected, this value corrupts every correlation, mean, and model coefficient in the pipeline. All cleaning steps are fully documented and reproducible via the accompanying Python notebook.